

The Prevalence of Phosphorus Containing Food Additives in Top Selling Foods in Grocery Stores

Objective To determine the prevalence of phosphorus-containing food additives in best selling processed grocery products and to compare the phosphorus content of a subset of top selling foods with and without phosphorus additives. **Design** The labels of 2394 best selling branded grocery products in northeast Ohio were reviewed for phosphorus additives. The top 5 best selling products containing phosphorus additives from each food category were matched with similar products without phosphorus additives and analyzed for phosphorus content. Four days of sample meals consisting of foods with and without phosphorus additives were created and daily phosphorus and pricing differentials were computed. **Setting** Northeast Ohio **Main outcome measures** Presence of phosphorus-containing food additives, phosphorus content **Results** 44% of the best selling grocery items contained phosphorus additives. The additives were particularly common in prepared frozen foods (72%), dry food mixes (70%), packaged meat (65%), bread & baked goods (57%), soup (54%), and yogurt (51%) categories. Phosphorus additive containing foods averaged 67 mg phosphorus/100 gm more than matched non-additive containing foods ($p=.03$). Sample meals comprised mostly of phosphorus additive-containing foods had 736 mg more phosphorus per day compared to meals consisting of only additive-free foods. Phosphorus additive-free meals cost an average of \$2.00 more per day. **Conclusion** Phosphorus additives are common in best selling processed groceries and contribute significantly to their phosphorus content. Moreover, phosphorus additive foods are less costly than phosphorus additive-free foods. As a result, persons with chronic kidney disease may purchase these popular low-cost groceries and unknowingly increase their intake of highly bioavailable phosphorus.